A Reduced-Order Axisymmetric Darcy Model for Espresso Extraction

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Abstract

This report summarizes the `Espresso Flow` repository, a reduced-order Python model for espresso extraction through a compressed porous coffee puck. The implementation combines a stochastic bimodal grind-size model, two-dimensional radial–axial heterogeneity, an axisymmetric Darcy pressure solve with an optional Forchheimer correction, and a simple two-pool extraction model driven by wetting and local flow speed. The goal is not to reproduce every machine-scale detail of espresso brewing, but to provide a compact and modifiable research code for sensitivity analysis, visualization, and rapid hypothesis testing.

1 Introduction

Espresso brewing can be interpreted as forced flow through a thin, highly compressed porous bed whose hydraulic resistance depends strongly on grind size, fines content, packing, and local heterogeneity. A useful model therefore has to do more than assign a single scalar permeability to the entire puck: it should allow the particle-size distribution to vary spatially, translate those variations into permeability, and then connect the resulting flow field to extraction dynamics. The repository described here follows exactly that strategy. It resolves the coffee bed as an axisymmetric r–z slice, draws local particle sizes from a bimodal distribution, computes an emergent permeability field, solves for pressure and velocity, and advances a reduced extraction model in time. The result is a lightweight numerical framework that sits between a toy lumped model and a fully resolved CFD simulation.

2 Technical background

2.1 Geometry and state variables

The puck is represented as the axisymmetric domain

$$\Omega = \{(r, z) : 0 \leq r \leq R,\ 0 \leq z \leq H\}, \tag{1}$$

where R is the basket radius and H is the puck depth. In the included baseline configuration, $R = 29\,\text{mm}$ and $H = 9\,\text{mm}$ on a 28×24 grid. Each cell stores local porosity $\phi(r, z)$, fines fraction $f(r, z)$, coarse-mode median particle size $d_c(r, z)$, an effective Sauter mean particle diameter $d_{32}(r, z)$, permeability $K(r, z)$, dry coffee mass, and the remaining fast and slow soluble masses.

The model separates the problem into three coupled layers:

1. stochastic bed construction,

2. hydraulic solution for pressure and velocity,
3. time-marching extraction and wetting dynamics.

This separation keeps the code compact while still allowing user control over grind settings, packing assumptions, and flow conditions.

2.2 Stochastic grind distribution and heterogeneous packing

At each cell, particle diameters are sampled from a bimodal lognormal mixture,

$$d \sim (1-f)\,\mathrm{LogNormal}(\ln d_c, \sigma_c^2) + f\,\mathrm{LogNormal}(\ln d_f, \sigma_f^2), \tag{2}$$

where d_c and d_f are the coarse and fines medians, σ_c and σ_f are log-space spreads, and f is the fines fraction. This is the main route by which the user controls the grind setting: the repository exposes the coarse median, fines median, fines fraction, and both lognormal widths directly in the JSON configuration and CLI.

Spatial heterogeneity is introduced by smooth Gaussian random fields. In the code, white noise is filtered in Fourier space so that the resulting perturbations have user-specified radial and axial correlation lengths. The local coarse median, fines fraction, and porosity are then defined as

$$d_c(r,z) = d_{c,0}\exp(\sigma_g\,\xi_g(r,z))\,, \tag{3}$$

$$f(r,z) = \mathrm{clip}\Big(f_0 + \sigma_f^{(h)}\xi_f(r,z),\ f_{\min},\ f_{\max}\Big)\,, \tag{4}$$

$$\phi(r,z) = \mathrm{clip}(\phi_u - 0.10C + \sigma_\phi\xi_\phi(r,z),\ \phi_{\min},\ \phi_{\max})\,, \tag{5}$$

where ϕ_u is the uncompressed porosity, C is the user-specified compression level, and ξ_g, ξ_f, and ξ_ϕ are approximately unit-variance Gaussian random fields. In other words, the compressed bed is not uniform; it contains structured radial–axial variation in particle size and porosity.

For each cell, the sampled particle diameters are reduced to a single effective size using the Sauter mean,

$$d_{32} = \frac{\sum_n d_n^3}{\sum_n d_n^2}. \tag{6}$$

This choice is natural for permeability closures because it weights particle size by surface-area relevance.

2.3 Permeability closure

The repository uses a Kozeny–Carman-style permeability law with an additional fines penalty,

$$K(r,z) = \alpha_K \frac{d_{32}(r,z)^2\,\phi(r,z)^3}{C_{KC}\,[1-\phi(r,z)]^2}\exp(-\gamma_f f(r,z))\,. \tag{7}$$

Here C_{KC} is the Kozeny–Carman constant, α_K is an overall calibration factor, and γ_f penalizes fines-rich regions. Equation (7) is deliberately semi-empirical. It preserves the expected increase of permeability with larger particles and larger void fraction, while still allowing the strong blocking effect of fines to be tuned directly. The implementation clips K to a physically reasonable range to prevent pathological cells from dominating the solve.

This closure is important conceptually: the user does not set permeability cell by cell. Instead, permeability emerges from grind statistics, compression, and heterogeneity. That makes it possible to ask physically meaningful questions such as: *what happens to shot time if the coarse mode shifts downward by* $20\,\mu\mathrm{m}$*, or if the fines fraction rises from 0.18 to 0.24?*

2.4 Axisymmetric Darcy flow and optional Forchheimer correction

Given the permeability field, the normalized pressure $\hat{p}(r, z)$ solves the steady axisymmetric porous-medium equation

$$\frac{1}{r}\frac{\partial}{\partial r}\left(rK\frac{\partial \hat{p}}{\partial r}\right) + \frac{\partial}{\partial z}\left(K\frac{\partial \hat{p}}{\partial z}\right) = 0. \tag{8}$$

The boundary conditions are

$$\hat{p}(r, 0) = 1, \qquad \hat{p}(r, H) = 0, \tag{9}$$

with zero radial flux at the symmetry axis and basket wall,

$$\frac{\partial \hat{p}}{\partial r}(0, z) = 0, \qquad \frac{\partial \hat{p}}{\partial r}(R, z) = 0. \tag{10}$$

A physical pressure drop is then imposed by

$$p(r, z, t) = \Delta P(t)\, \hat{p}(r, z), \tag{11}$$

where the baseline schedule uses 2 bar for 6 s of preinfusion and 9 bar afterward.

The Darcy velocity is

$$\mathbf{u}_D = -\frac{K}{\mu}\nabla p, \tag{12}$$

with fluid viscosity μ. The repository also offers a local Forchheimer-style correction for higher gradients. Rather than solving a fully coupled nonlinear pressure equation, it rescales the Darcy velocity magnitude using

$$\|\nabla p\| = \frac{\mu}{K}u + \beta\frac{\rho}{\sqrt{K}}u^2, \tag{13}$$

where $u = \|\mathbf{u}\|$, ρ is fluid density, and β is a user-controlled inertial parameter. Solving the quadratic for the positive root gives

$$u = \frac{-\mu/K + \sqrt{(\mu/K)^2 + 4\beta\rho\|\nabla p\|/\sqrt{K}}}{2\beta\rho/\sqrt{K}}, \tag{14}$$

while the direction is inherited from the Darcy vector $\mathbf{u}_D$. This is a pragmatic compromise: it captures some inertial suppression of flow without requiring a nonlinear solve at each time step.

The outlet flow rate is approximated from the bottom-row axial velocity and annular cell areas,

$$Q_{\text{out}} \approx \sum_i u_z(r_i, H)\, \pi \left[\left(r_i + \frac{\Delta r}{2}\right)^2 - \left(r_i - \frac{\Delta r}{2}\right)^2\right]. \tag{15}$$

Because the permeability field is fixed over a run in the current version, the hydraulic response is piecewise steady within each pressure stage; most temporal structure then comes from wetting and depletion of soluble material rather than from evolving resistance.

2.5 Wetting, solute release, and extraction metrics

Each cell is initialized with dry coffee mass proportional to its solid volume fraction, then split into fast and slow soluble pools,

$$M_{\text{sol}} = \chi_s M_{\text{dry}}, \qquad M_f = \chi_f M_{\text{sol}}, \qquad M_s = (1 - \chi_f) M_{\text{sol}}, \tag{16}$$

where χ_s is the total soluble fraction and χ_f is the fast-pool fraction. The wetting state $w(r, z, t)$ evolves heuristically as

$$w^{n+1} = \mathrm{clip}\left(w^n + \Delta t \, \frac{\|\mathbf{u}\|}{u_w},\ 0,\ 1 \right), \tag{17}$$

so high-flow regions wet more quickly.

Local release rates are then written as

$$k_f = k_{f,0} \, \exp[\eta_T (T - T_{\mathrm{ref}})] \, w \left(\frac{\|\mathbf{u}\|}{u_{\mathrm{ref}}} \right)^{a_f}, \tag{18}$$

$$k_s = k_{s,0} \, \exp[\eta_T (T - T_{\mathrm{ref}})] \, w \left(\frac{\|\mathbf{u}\|}{u_{\mathrm{ref}}} \right)^{a_s}, \tag{19}$$

with different baseline rates and velocity exponents for the fast and slow pools. Over one time step,

$$\Delta M_f = M_f^n \left(1 - e^{-k_f \Delta t} \right), \tag{20}$$

$$\Delta M_s = M_s^n \left(1 - e^{-k_s \Delta t} \right). \tag{21}$$

The beverage mass is updated by

$$m_b^{n+1} = m_b^n + \rho Q_{\mathrm{out}} \Delta t, \tag{22}$$

and the cumulative extracted solids by

$$m_{\mathrm{sol}}^{n+1} = m_{\mathrm{sol}}^n + \sum_{i,j} (\Delta M_f + \Delta M_s)_{ij}. \tag{23}$$

The reported metrics follow directly:

$$\mathrm{TDS}_{\mathrm{inst}} = 100 \frac{\Delta m_{\mathrm{sol}}}{\Delta m_b}, \tag{24}$$

$$\mathrm{TDS}_{\mathrm{cum}} = 100 \frac{m_{\mathrm{sol}}}{m_b}, \tag{25}$$

$$\mathrm{EY} = 100 \frac{m_{\mathrm{sol}}}{m_{\mathrm{dose}}}. \tag{26}$$

This reduced chemistry is intentionally simple, but it is expressive enough to produce sensible early-shot TDS peaks, cumulative extraction growth, and spatial extraction maps that reflect preferential flow pathways.

2.6 Numerical structure

The pressure equation is discretized on the axisymmetric grid using harmonic means of permeability at cell faces, which improves flux continuity across sharp heterogeneity. The resulting linear system is solved directly with dense linear algebra. Time integration of wetting and extraction is explicit, using a fixed step size Δt. For uncertainty studies, the repository repeats the entire pipeline over multiple random seeds.

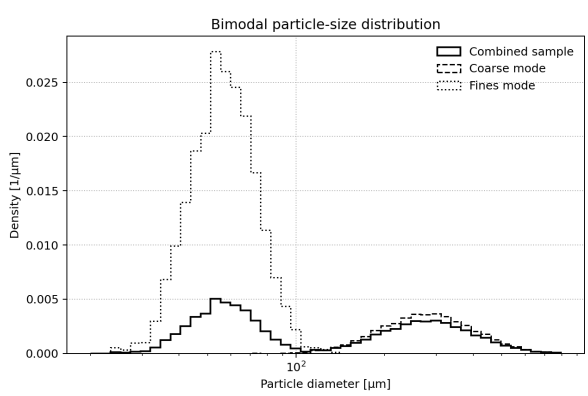
3 Example results from the bundled demo outputs

The included baseline example uses an 18 g dose, coarse median size 320 µm, fines median size 60 µm, fines fraction 0.18, baseline brew water at 93 °C, 2 bar preinfusion for 6 s, and 9 bar brewing to 32 s. The saved summary reports a target beverage mass of 36 g reached at 31.6 s, a final extraction yield of 21.38%, a final cumulative TDS of 10.47%, a mean porosity of 0.375, and a median permeability of $1.72 \times 10^{-15}\,\mathrm{m}^2$.

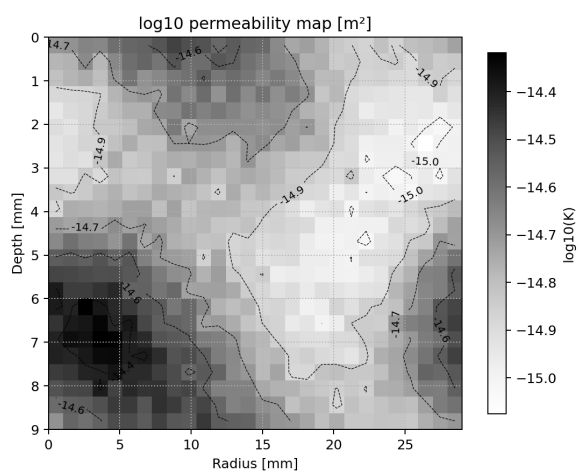
A six-run Monte Carlo demonstration is also included. Across those runs, the mean time to the 36 g target is 28.77 ± 3.00 s, the mean final extraction yield is 21.60 ± 0.15%, and the mean final cumulative TDS is 9.49%. This is a useful reminder that even a fixed nominal recipe produces a spread in outcomes once stochastic bed heterogeneity is admitted.

Table 1: Selected metrics from the included demo outputs.

Metric	Baseline demo	Monte Carlo demo
Dose [g]	18.0	18.0
Target beverage mass [g]	36.0	36.0
Time to target mass [s]	31.6	28.77 ± 3.00
Final extraction yield [%]	21.38	21.60 ± 0.15
Final cumulative TDS [%]	10.47	9.49 (mean)
Median permeability [m^2]	1.72×10^{-15}	$(1.82 \pm 0.07) \times 10^{-15}$
Mean porosity [-]	0.375	—

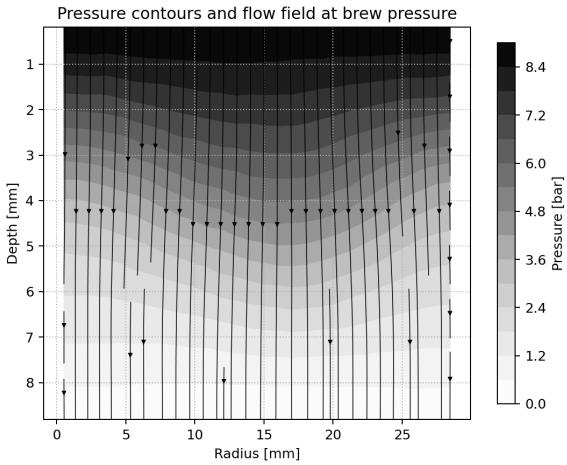


(a) Sampled bimodal particle-size distribution.

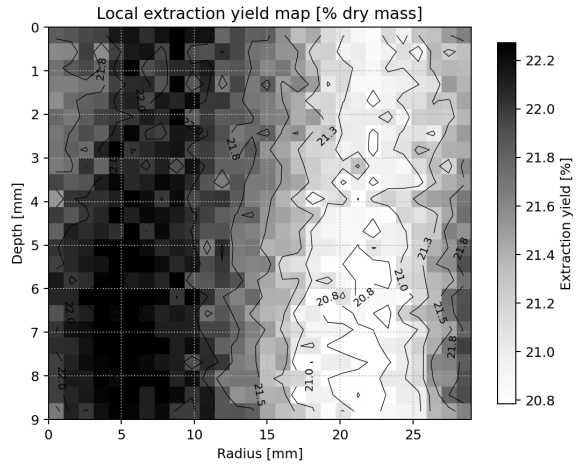


(b) Resulting heterogeneous permeability field.

Figure 1: Bed construction outputs from the bundled baseline run. The particle-size distribution is explicitly bimodal, and the permeability map shows that stochastic grind and packing heterogeneity generate large-scale radial–axial structure rather than uniform resistance.



(a) Pressure contours and flow streamlines.



(b) Final local extraction-yield map.

Figure 2: Hydraulic and extraction fields from the baseline run. The flow field is predominantly axial, but permeability variations bend streamlines and create regions of slightly elevated or suppressed extraction.

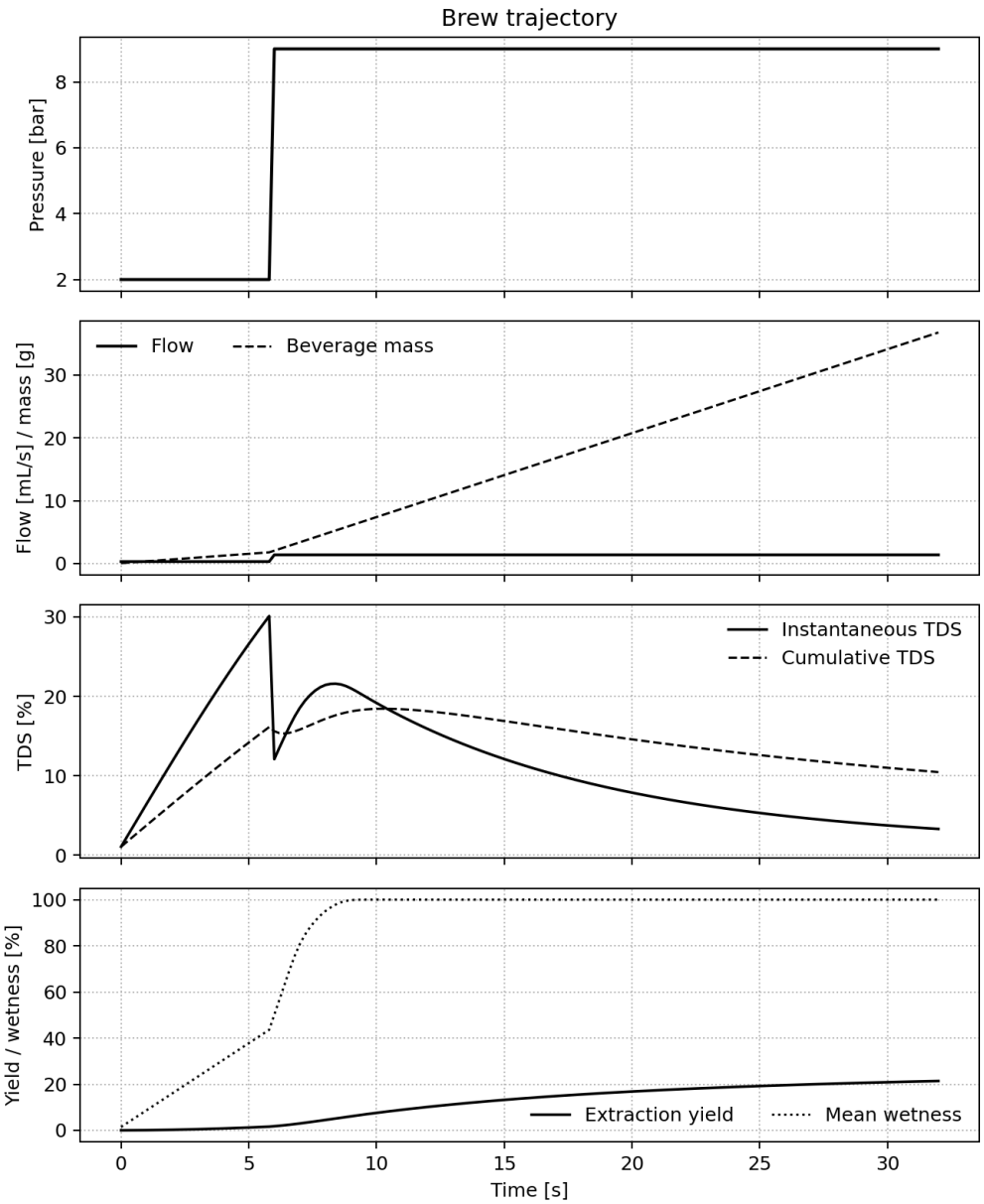


Figure 3: Time-series outputs from the baseline simulation. The pressure step at 6 s separates pre-infusion from full brewing. Because the hydraulic field is fixed within each pressure stage, the flow curve is piecewise steady, while TDS, wetting, beverage mass, and extraction evolve dynamically.

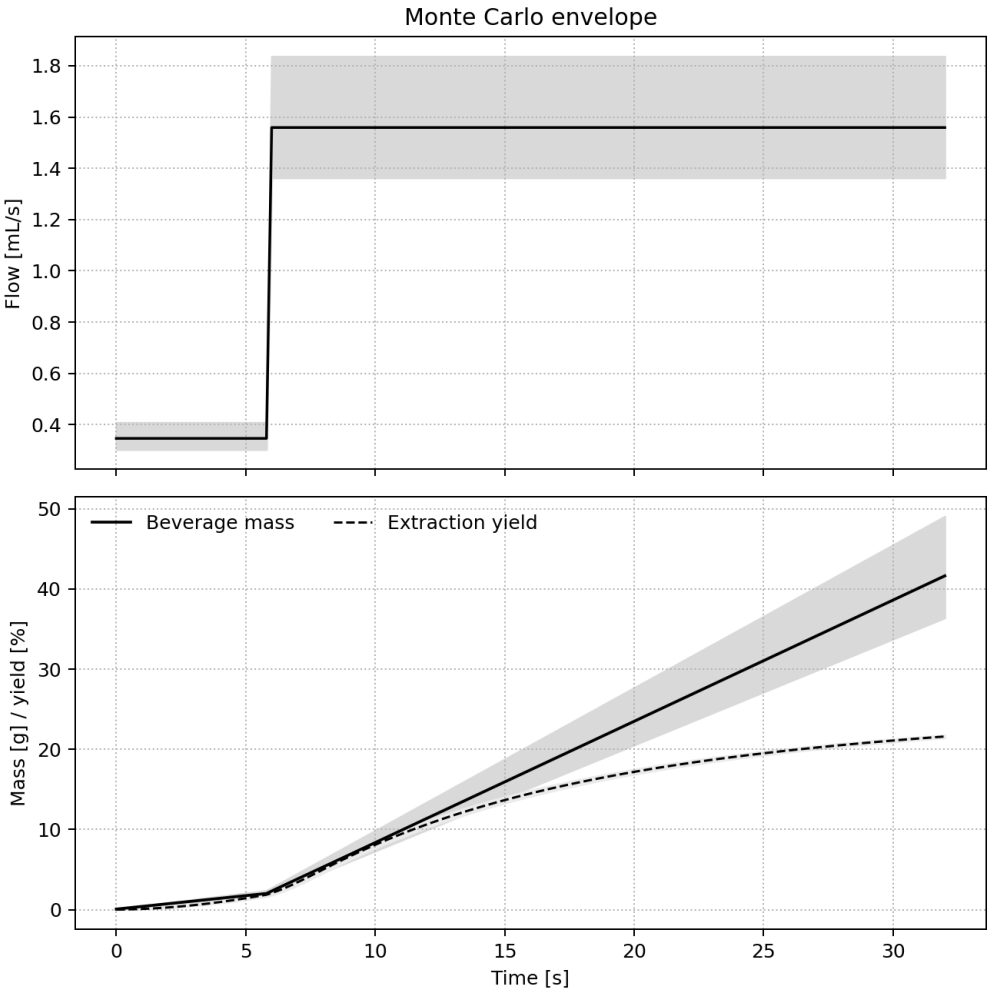


Figure 4: Monte Carlo envelope from the bundled uncertainty example. Shaded bands show the spread induced by stochastic bed realization across repeated runs with the same nominal recipe.

4 Discussion

The repository demonstrates that a relatively small amount of structure goes a long way in espresso modeling. Once the puck is given a spatially varying particle-size distribution and a permeability law derived from that distribution, the resulting pressure and velocity fields naturally produce nonuniform extraction. The model also cleanly separates user-facing controls from derived fields: changing grind medians, fines fraction, compression, or permeability scale propagates through to flow rate, shot time, and extraction yield without the user having to edit the numerical solver.

At the same time, the current implementation remains deliberately reduced-order. The wetting law is heuristic, the Forchheimer correction is local rather than fully coupled, and the bed does not evolve hydraulically during the shot. In particular, fines migration, swelling, erosion, fracture, and unsaturated infiltration are not yet resolved. For that reason, the repository is best viewed as a research scaffold and visualization tool rather than a validated digital twin of a specific machine or coffee.

5 Conclusion

This Python repository provides a practical middle ground between simplicity and physical structure. It includes the essential ingredients needed for exploratory espresso simulations: a user-controlled bimodal grind model, 2D heterogeneous bed generation, Darcy-based flow, an optional inertial correction, reduced extraction kinetics, and black-and-white figures suitable for reports. As a foundation, it is already useful for sensitivity analysis and uncertainty quantification; as a codebase, it is well positioned for extension toward fines migration, saturation dynamics, thermal transport, and data-driven calibration.

Software citation

Software metadata for this repository are included in `CITATION.cff`. A concise citation is: `Bailey, M. M. (2026). Espresso Flow (Version 0.1.0) [Computer software]`.